

**A REPORT ON
5(FIVE) DAYS [18.07.22-22.07.22] FACULTY DEVELOPMENT
PROGRAM (ONLINE) ON
INTRODUCTION TO DATA ANALYTICS USING MATLAB**

A five days (05) Faculty Development Program has been successfully conducted from 18th - 22nd July, 2022 by the **Dept. of Electrical Engineering, Institute of Engineering & Management, Kolkata** in collaboration with **National Institute of Technical Teachers' Training & Research (NITTTR), Kolkata [MHRD]**. The objective of the FDP is to provide a platform for discussion on computational intelligence on the basis of MATLAB. The main motive behind the program is to learn and implement the computational intelligence in research area. The resource persons were invited from the eminent Institute NITTTR, Kolkata. Faculties from different Institutes of India along with the IEM-UEM family members were invited and among them around 85 were actively participated to this program. The said program was fully coordinated by Dr. Mandakinee Bandyopadhyay (Asst. Professor, Dept. of Electrical Engineering, Institute of Engineering & Management, Kolkata).

Summary of the said program is as follows:

Day 1: In the first half, Dr. N. K. Mandal introduced the importance of Data analysis in our daily life. Then Dr. Mandal directed about hands-on training on MATLAB fundamentals by showing the basic Matlab operations like array operations, matrix operations, function creation, and plotting.

In the second half, Dr. Mandal discussed about the linear regression technique for single and multiple independent variables using the least-square method.

Day 2: In the first half, Dr. C. Chakraborty delivered a lecture on Basic Statistics including mean, median, mode, standard deviation, and linear regression, and concluded his lecture with logistic regression.

In the second half, the Optimization technique for various engineering problems has been discussed by Dr. N. Mandal along with some hands on practice using MATLAB

Day 3: In the first half, Dr. I. Saha delivered his lecture on Supervised and Unsupervised learning. Dr. Saha emphasized the method of data clustering using the k-means algorithm.

In the second half, Dr. Saha taught about hands-on training on clustering using some known datasets followed by quizzes on the whole day lecture. The goodness of fit of the clustering has also been checked using a well-known method.

Day 4: In the first half, Dr. S. Roy delivered a lecture on the application of Machine Learning in real life by showing some real life case studies like classification and visualization of Alzheimer's Disease like volumetric convolution neural network and transfer learning.

In the second half, Dr. C. Chakraborty gave a lecture on Data Classification using the Bayesian method. Then he solved an air traffic problem using the Naive Bayes method. Finally, Dr. Chakraborty concluded his lecture with Support Vector Machine (SVM).

Day 5: In the first half 1st half, Dr. N. K. Mandal delivered a lecture on Monte carlo simulation and markov chain.

In the 2nd half, the Principal Component Analysis was described followed by the hands on practice using MATLAB.

The program was concluded on 22nd July by the valedictory session in presence of the members of the advisory committee and the resource persons.

Here are some glimpses of the program:

